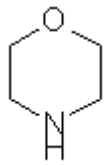
Morpholine

Technical Data Sheet
July 2009

Formula..... C_2H_7NO
Molecular Weight 87.1
PRD Number 30036734
CAS Registry Number 110-91-8



Product Specifications	Value	Test Method
Assay, % minimum	99.0	GC
Water, % maximum	0.30	DIN 51 777
Color, APHA maximum	20	DIN EN 1557
Foreign matter	None	STI 8412

Physical Properties	
Freezing point, ºC	-5
Flash point, ºC	32
Boiling point ºC	128-130
Density @ 20ºC, g/cm^3	1.016
Ignition temperature ºC	275

Description
Morpholine is a colorless to yellow liquid with a perceptible amine-like odor. It is miscible with water.

Synonyms
Tetrahydro-1,4-oxazine

Safety
Morpholine is irritating to skin, and harmful if swallowed. There is risk to serious damage to eyes and to health if prolonged exposure. It is also flammable. Wear protective clothing, gloves, and eye/face protection.

Always refer to the Material Safety Data Sheet (MSDS) for detailed information on safety.

Applications
Intermediate used in the production of:
- Corrosion inhibitor
- Crop protection agents
- Dyes
- Optical brighteners
- Pharmaceuticals
- Rubber chemicals

Packaging
This product is available in bulk and drums.

Storage & Handling
Morpholine has a shelf life of 24 months. It must be protected from fire, explosions, sources of ignition and electrostatic charges, acids and acid forming substances. Ensure thorough ventilation of work and storage area.

Always refer to the Material Safety Data Sheet (MSDS) for detailed information on handling and disposal.

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Contact us:

E-mail: chemical_intermediates @ basf-corp.com
Website: www. basf .com /usa/intermediates

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